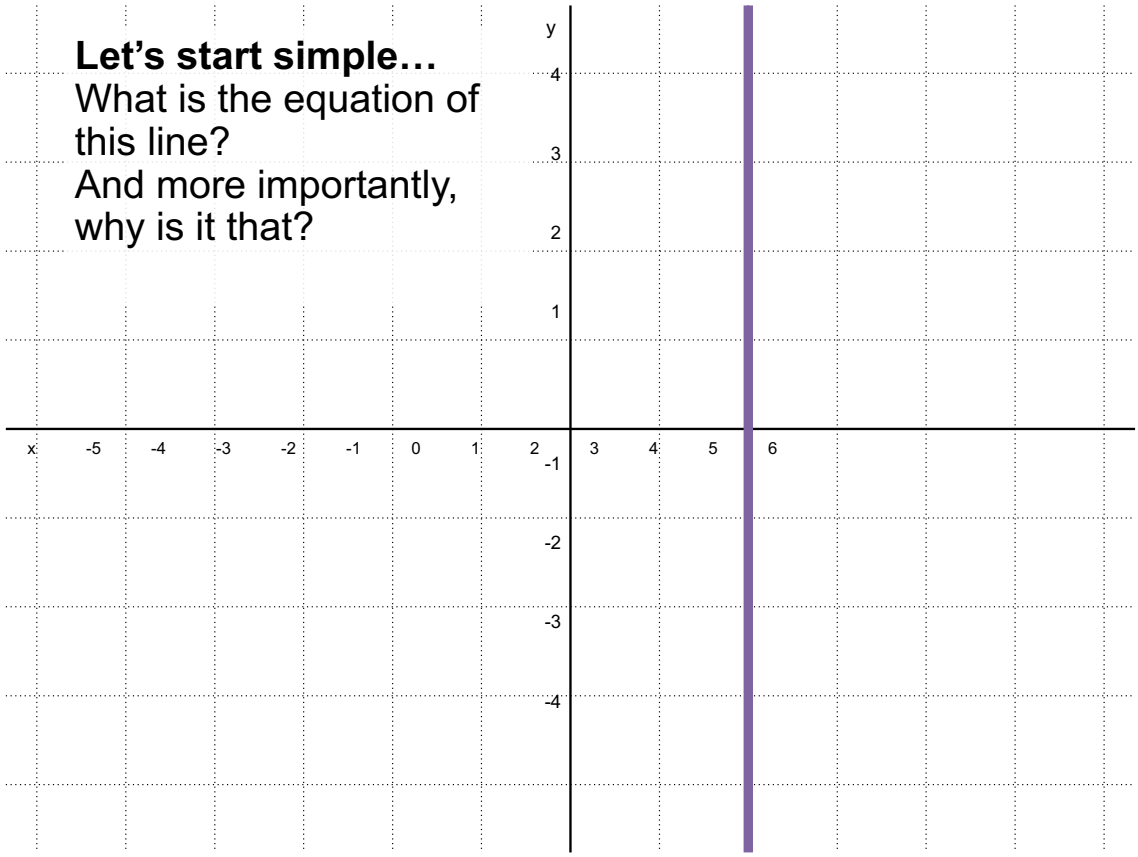
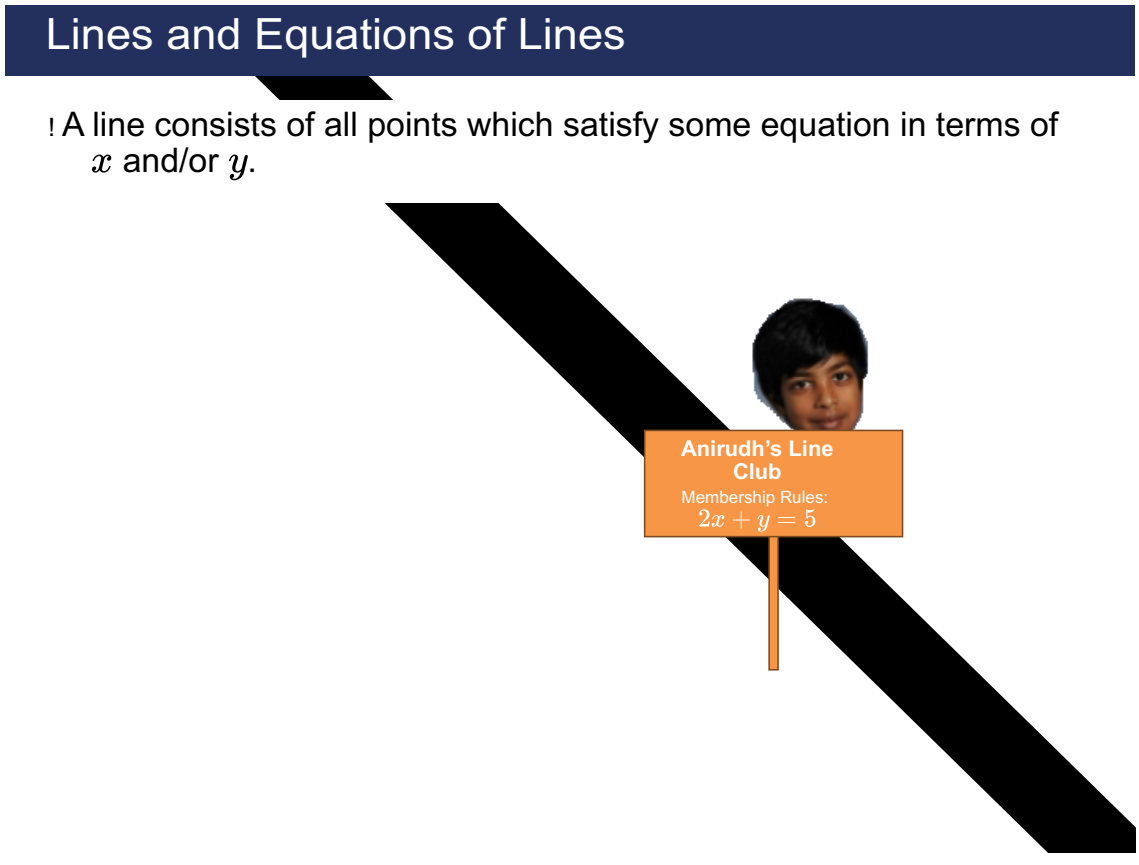
1.03 Coordinate Geometry: Linear Graphs

Chapter 5 · Original slide 4



Chapter 5 · Original slide 5



1.03 Coordinate Geometry: Linear Graphs

Chapter 5 · Original slide 6

Examples

This means we can **substitute** the values of a coordinate into our equation whenever we know the point lies on the line.

The point $(5, a)$ lies on the line with equation $y = 3x + 2$. Determine the value of a.

Find the coordinate of the point where the line $2x + y = 5$ cuts the x-axis.

Chapter 5 · Original slide 7

Test Your Understanding

Determine where the line $x + 2y = 3$ crosses the:

a) y-axis:

b) x-axis:

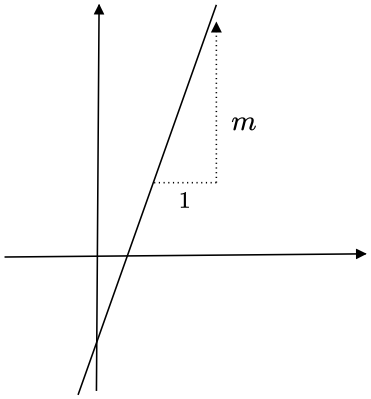
What mistakes do you think it's easy to make?

1.03 Coordinate Geometry: Linear Graphs

Chapter 5 · Original slide 8

Recap of gradient

The steepness of a line is known as the **gradient**.
It tells us what y changes by as x increases by 1.



So if the y value increased by 6 as the x value increased by 2, what is y increasing by for each unit increase of x?
How would that give us a suitable formula for the gradient m?

$$m = \frac{\Delta y}{\Delta x}$$

Chapter 5 · Original slide 9

Examples

Find the gradient of the line that goes through the points:

1. $(1, 4)$ $(3, 10)$
2. $(5, 7)$ $(8, 1)$
3. $(2, 2)$ $(-1, 10)$
4. Show that the points $A(3, 4), B(5, 5), C(11, 8)$ all lie on a straight line.

If points are ‘collinear’ they lie on the same line.

1.03 Coordinate Geometry: Linear Graphs

Chapter 5 · Original slide 10

Further Example

The line joining $(2, -5)$ to $(4, a)$ has gradient -1. Work out the value of a.

Chapter 5 · Original slide 11

$y = mx + c$

One form we can put a straight line equation in is:

$$y = mx + c$$

Gradient

y-intercept

Why does it work?

- **The $\mathbf{y}$-intercept by definition is the $\mathbf{y}$ value when $\mathbf{x=0}$.**
  Substituting:
  $\mathbf{y=(m\times 0)+c=c}$ **as expected.**
- **By the definition of gradient, if we increase $\mathbf{x}$ by 1, $\mathbf{y}$ should increase by $\mathbf{m}$:**
  $\mathbf{y=m\,(x+1)+c=mx+m+c}$
  which indeed has increased by $\mathbf{m}$.

1.03 Coordinate Geometry: Linear Graphs

Chapter 5 · Original slide 12

$y = mx + c$

Determine the gradient and y-intercept of the line with equation $4x - 3y + 5 = 0$

Make y the subject so we have the form $y = mx + c$
Put y on the side it's positive.

Chapter 5 · Original slide 13

$ax + by + c = 0$

At GCSE, $\mathbf{y=mx+c}$ was the main form you would express a straight line equation, sometimes known as the '**slope-intercept form**'.
But another common form is $\mathbf{ax+by+c=0}$, where a, b, c are integers.
This is known as the '**standard**' form.

Express $y = \frac{1}{3}x - \frac{2}{3}$ in the form $ax + by + c = 0$, where a, b, c are integers.

We don't want fractions, so multiply by an appropriate number.

Put everything on either side of equation.
$-x + 3y + 2 = 0$ would also be OK.

We'll see on the next slide WHY we might want to put an equation in this form over $y = mx + c$...

1.03 Coordinate Geometry: Linear Graphs

Chapter 5 · Original slide 14

Just for your interest…

Why might we want to put a straight line equation in the form $ax + by + c = 0$?

$y = mx + c$
"Slope-Intercept Form"

$ax + by + c = 0$
"Standard Form"

Coverage

$y = mx + c$ doesn't allow you to represent vertical lines. Standard form allows us to do this by just making b zero.

$x + 4 = 0$

Symmetry

In general, the '**linear combination**' of two variables x and y is $ax + by$, i.e. "some amount of x and some amount of y". There is a greater elegance and symmetry to this form over $y = mx + c$ because x and y appear similarly within the expression.

Usefulness

This more 'elegant' form also means it ties in with vectors and matrices. In FM, you will learn about the '**dot product**' of two vectors:

$$\begin{pmatrix} a \\ b \end{pmatrix} \cdot \begin{pmatrix} x \\ y \end{pmatrix} = ax + by$$

thus since $ax + by + c = 0$, we can represent a straight line using:

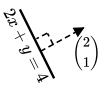
$$\begin{pmatrix} a \\ b \end{pmatrix} \cdot \begin{pmatrix} x \\ y \end{pmatrix} + c = 0 \qquad (1)$$

We can extend to 3D points to get the equation of a **plane**:

$$\begin{pmatrix} a \\ b \\ c \end{pmatrix} \cdot \begin{pmatrix} x \\ y \\ z \end{pmatrix} + d = 0 \qquad (2)$$

Conveniently, in equation (1), the vector $\begin{pmatrix} a \\ b \end{pmatrix}$ is **perpendicular to the line**. And in equation (2), the vector $\begin{pmatrix} a \\ b \\ c \end{pmatrix}$ is perpendicular to the plane. Nice!

$2x + y = 4$ ➡ $\begin{pmatrix} 2 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} x \\ y \end{pmatrix} = 4$ ➡



Chapter 5 · Original slide 15

Test Your Understanding

Express $y = \frac{2}{5}x + \frac{3}{5}$ in the form $ax + by + c = 0$, where a, b, c are integers.

1.03 Coordinate Geometry: Linear Graphs

Chapter 5 · Original slide 16

Exercise 5A – Pure Y1 Page 91

9 Show that the points $A(2, 3)$, $B(4, 4)$ and $C(10, 7)$ can be joined by a straight line.

10 Show that the points $A(-2a, 5a)$, $B(0, 4a)$ and points $C(6a, a)$ are collinear. $(3\ \mathrm{ma}\ldots$

Chapter 5 · Original slide 17

Exercise 5B – Pure Y1 Page 92-93

8 A line l cuts the x-axis at $(5, 0)$ and the y-axis at $(0, 2)$.

a Find the gradient of the line.

b Find an equation of the line in the form $ax + by + c = 0$.

9 Show that the line with equation $ax + by + c = 0$ has gradient $-\frac{a}{b}$ and cuts the y-axis at $-\frac{c}{b}$.

10 The line l with gradient 3 and y-intercept $(0, 5)$ has the equation $ax - 2y + c = 0$.

Find the values of a and c. (2 marks)

1.03 Coordinate Geometry: Linear Graphs

Chapter 5 · Original slide 18

Exercise 5B – Pure Y1 Page 92-93

11 The straight line l passes through $(0, 6)$ and has gradient -2. It intersects the line with equation $5x - 8y - 15 = 0$ at point P. Find the coordinates of P.

12 The straight line l_1 with equation $y = 3x - 7$ intersects the straight line l_2 with equation $ax + 4y - 17 = 0$ at the point $P(-3, b)$.

a Find the value of b.

b Find the value of a.

Chapter 5 · Original slide 19

Equations using two points/point + gradient

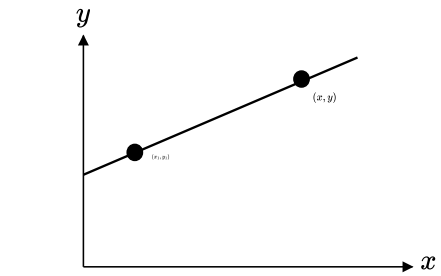
Find the equation of the line that goes through $(3, 5)$ and has gradient 2.

How would you have done this at **GCSE**?

1.03 Coordinate Geometry: Linear Graphs

Chapter 5 · Original slide 20

A new way…



Notes: Note that x_1 and y_1 are constants while x and y are variables. The latter are variables because as these 'vary', we get different points on the line.

Suppose that (x_1, y_1) is some fixed point on the line that we specify (e.g. $(3, 5)$).
Suppose that (x, y) represents a generic point on the line, which is allowed to change as we consider different points on this line.
Then:

$$m = \frac{y-y_1}{x-x_1}$$

Thus:

Chapter 5 · Original slide 21

Quickfire Questions

In a nutshell: You can use this formula whenever you have (a) a gradient and (b) any point on the line.

Gradient	Point	(Unsimplified) Equation
3	$(1, 2)$	
5	$(3, 0)$	
2	$(-3, 4)$	
$\frac{1}{2}$	$(1, -5)$	
9	$(-4, -4)$	

1.03 Coordinate Geometry: Linear Graphs

Chapter 5 · Original slide 22

Using 2 points

Find the equation of the line that goes through $(4, 5)$ and $(6, 2)$, giving your equation in the form $ax + by + c = 0$.

We need to work out the gradient first before we can use $y - y_1 = m(x - x_1)$:

Test Your Understanding:
Find the equation of the line that goes through $(-1, 9)$ and $(4, 5)$, giving your equation in the form $ax + by + c = 0$.

Chapter 5 · Original slide 23

Exercise 5C – Pure Y1 Page 94-95

3 Find the equation of the line l which passes through the points $A(7, 2)$ and $B(9, -8)$. Give your answer in the form $ax + by + c = 0$.

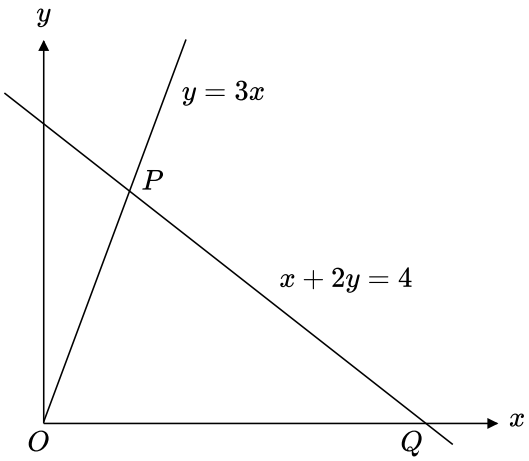
5 The straight line l passes through $(a, 4)$ and $(3a, 3)$. An equation of l is $x + 6y + c = 0$. Find the value of a and the value of c.

6 The straight line l passes through $(7a, 5)$ and $(3a, 3)$. An equation of l is $x + by - 12 = 0$. Find the value of a and the value of b.

1.03 Coordinate Geometry: Linear Graphs

Chapter 5 · Original slide 24

Intersection of lines



The diagram shows two lines with equations $y = 3x$ and $x + 2y = 4$, which intersect at the point P.

a) Determine the coordinates of P.

We know from Chapter 4 we can just solve two equations simultaneously.

b) The line $x + 2y = 4$ intersects the x-axis at the point Q. Determine the coordinate of Q.

Chapter 5 · Original slide 25

Test Your Understanding

C1 Edexcel May 2013 Q6

The straight line L_1 passes through the points $(-1, 3)$ and $(11, 12)$.
(a) Find an equation for L_1 in the form $ax + by + c = 0$, where a, b and c are integers.
\hfill(4)
The line L_2 has equation $3y + 4x - 30 = 0$.
(b) Find the coordinates of the point of intersection of L_1 and L_2. \hfill(3)

(a)

(b)

1.03 Coordinate Geometry: Linear Graphs

Chapter 5 · Original slide 26

Exercise 5D – Pure Y1 Page 96

6 The line that passes through the points $(-3, -5)$ and $(4, 9)$ meets the y-axis at the point G. Work out the coordinates of the point G.

9 The lines $y = 4x - 10$ and $y = x - 1$ intersect at the point T. Find the equation of the line with gradient $-\frac{2}{3}$ that passes through the point T. Write your answer in the form $ax + by + c = 0$, where a, b and c are integers.

Chapter 5 · Original slide 27

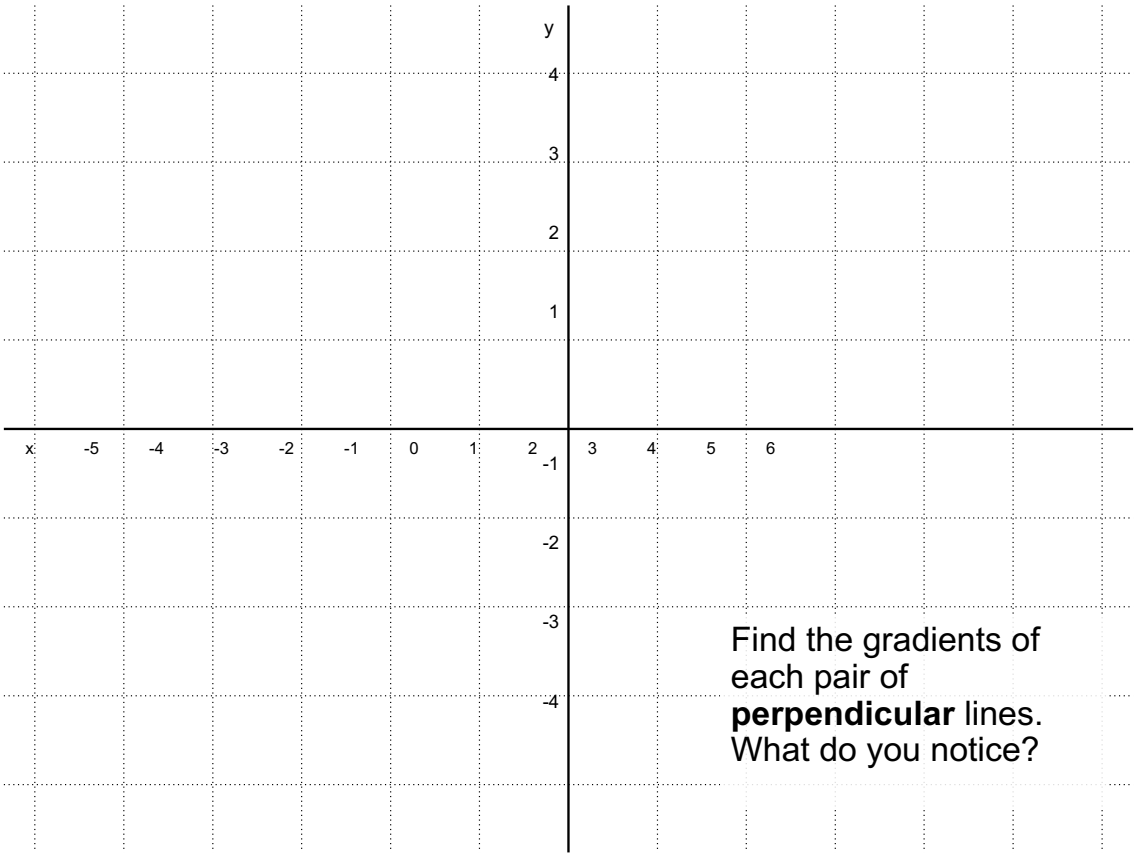
Exercise 5D – Pure Y1 Page 96

14 The line $y = 4x + 5$ meets the y-axis at the point C. The line $y = -3x - 15$ meets the x-axis at the point D. Find the equation of the line joining the points C and D. Write your answer in the form $ax + by + c = 0$, where a, b and c are integers.

16 The lines $y = -2x + 1$ and $y = x + 7$ intersect at the point L. The point M has coordinates $(-3, 1)$. Find the equation of the line that passes through the points L and M.

1.03 Coordinate Geometry: Linear Graphs

Chapter 5 · Original slide 28



Chapter 5 · Original slide 29

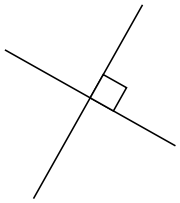
Perpendicular Lines

The gradients of parallel lines are equal.
If two lines are perpendicular, then the gradient of one is the **negative reciprocal** of the other.

$$m_1 = -\frac{1}{m_2}$$

To **show** that two lines are perpendicular:

$$m_1 m_2 = -1$$



Gradient	Gradient of Perpendicular Line
2	
-3	
$\frac{1}{4}$	
5	
$-\frac{2}{7}$	
$\frac{7}{5}$	

1.03 Coordinate Geometry: Linear Graphs

Chapter 5 · Original slide 30

Example Problems

1. A line is goes through the point (9,10) and is perpendicular to another line with equation $y = 3x + 2$. What is the equation of the line?

2. A line L_1 goes through the points $A(1, 3)$ and $B(3, -1)$. A second line L_2 is perpendicular to L_1 and passes through point B. Where does L_2 cross the x-axis?

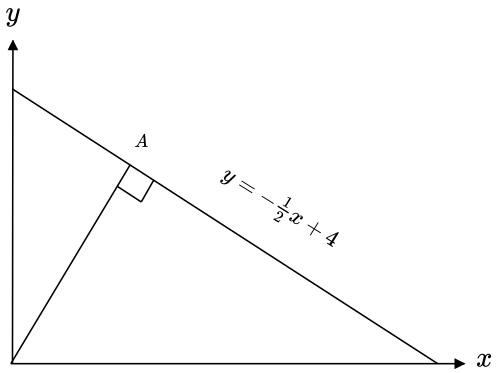
3. Are the following lines parallel, perpendicular, or neither?
$y = \frac{1}{2}x$
$2x - y + 4 = 0$

Chapter 5 · Original slide 31

Test Your Understanding

1. A line goes through the point (4,7) and is perpendicular to another line with equation y=2x+2. What is the equation of the line? Put your answer in the form ax+by+c=0, where a,b,c are integers.

2. Determine the point A.



1.03 Coordinate Geometry: Linear Graphs

Chapter 5 · Original slide 32

Exercise 5E/5F – Pure Y1 Page 96

3 The coordinates of a quadrilateral $ABCD$ are $A(-6, 2)$, $B(4, 8)$, $C(6, 1)$ and $D(-9, -8)$. Show that the quadrilateral is a trapezium.

7 A line is parallel to the line $2x - 3y - 1 = 0$ and it passes through the point $(0, 0)$. Write down the equation of the line.

Chapter 5 · Original slide 33

Exercise 5F – Pure Y1 Page 100

9 The line l passes through the points $(-3, 0)$ and $(3, -2)$ and the line n passes through the points $(1, 8)$ and $(-1, 2)$. Show that the lines l and n are perpendicular.

10 The vertices of a quadrilateral $ABCD$ have coordinates $A(-1, 5)$, $B(7, 1)$, $C(5, -3)$ and $D(-3, 1)$. Show that the quadrilateral is a rectangle.

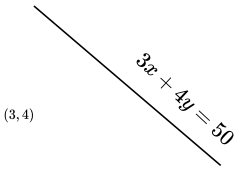
1.03 Coordinate Geometry: Linear Graphs

Chapter 5 · Original slide 34

Exercise 5E/5F – Pure Y1 Page 96

Extension Problems

- [MAT 2004 1D]
  What is the reflection of the point $(3, 4)$ in the line $3x + 4y = 50$?

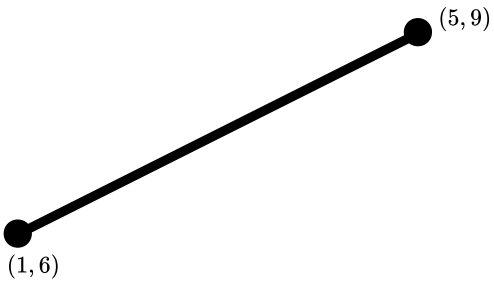


- [MAT 2014 1D] The reflection of the point $(1, 0)$ in the line $y = mx$ has coordinates: (in terms of m)

- *[STEP I 2004 Q6]* The three points A, B, C have coordinates $(p_1, q_1), (p_2, q_2)$ and (p_3, q_3), respectively. Find the point of intersection of the line joining A to the midpoint of BC, and the line joining B to the midpoint of AC. Verify that this point lies on the line joining C to the midpoint of AB.
  The point H has coordinates $(p_1 + p_2 + p_3, q_1 + q_2 + q_3)$. Show that if the line AH intersects the line BC at right angles, then $p_2^2 + q_2^2 = p_3^2 + q_3^2$, and write down a similar result if the line BH intersects the line AC at right angles.
  Deduce that if AH is perpendicular to BC and also BH is perpendicular to AC, then CH is perpendicular to AB.

Chapter 5 · Original slide 35

Distances between points



How could we find the **distance** between these two points?

Hint: Pythagoras

Form a right-angled triangle using the change in x and change in y, then use Pythagoras.

1.03 Coordinate Geometry: Linear Graphs

Chapter 5 · Original slide 36

Examples

Distance between:

(3,4) and $(5, 7)$
(5,1) and $(6, -3)$
(0,-2) and $(-1, 3)$

Note: Unlike with gradient, we don't care if the difference is positive or negative (it's being squared to make it positive anyway!)

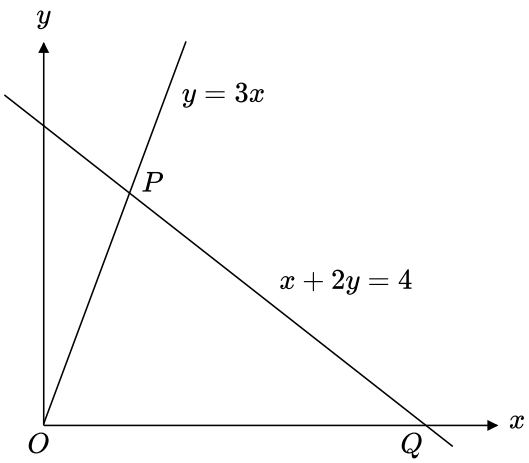
Quickfire Questions:

Distance between:

(1,10) and $(4, 14)$
(3,-1) and $(0, 1)$
(-4,-2) and $(-12, 4)$

Chapter 5 · Original slide 37

Area of Shapes



The diagram shows two lines with equations $y = 3x$ and $x + 2y = 4$, which intersect at the point P.

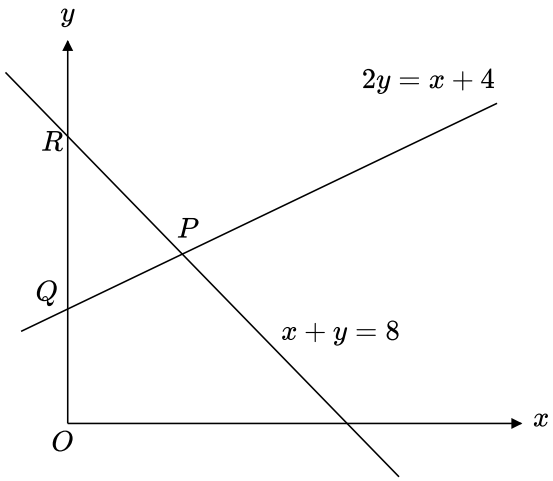
a) Determine the coordinates of P. (We did this in a previous lesson)

b) The line $x + 2y = 4$ intersects the x-axis at the point Q. Determine the area of the triangle OPQ.

1.03 Coordinate Geometry: Linear Graphs

Chapter 5 · Original slide 38

Further Example



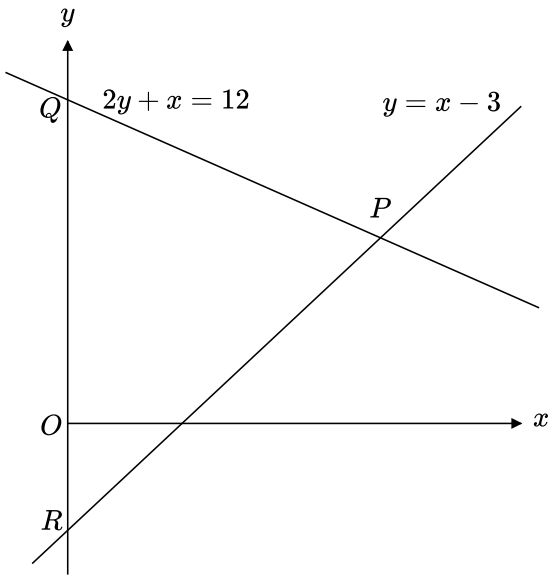
Tip: When finding areas of triangles in exam questions, one line is often vertical or horizontal. You should generally choose this to be the 'base' of your triangle.

a Determine the length of PQ.

b Determine the area PQR.

Chapter 5 · Original slide 39

Test Your Understanding



a) Determine the coordinate of P.

b) Determine the area of PQR.

c) Determine the length PQ.

1.03 Coordinate Geometry: Linear Graphs

Chapter 5 · Original slide 40

Exercise 5G – Pure Y1 Page 102

7 A point P lies on the line with equation $y = 4 - 3x$. The point P is a distance $\sqrt{34}$ from the origin. Find the two possible positions of point P. \hfill$(5\ \mathrm{ma}\ldots$

8 The vertices of a triangle are $A(2, 7)$, $B(5, -6)$ and $C(8, -6)$.
a Show that the triangle is a scalene triangle.
b Find the area of the triangle ABC.

Chapter 5 · Original slide 41

Exercise 5G – Pure Y1 Page 103

12 The straight line l_1 passes through the point $(-4, 14)$ and has gradient $-\frac{1}{4}$.
a Find an equation for l_1 in the form $ax + by + c = 0$, where a, b and c are integers.
b Write down the coordinates of A, the point where straight line l_1 crosses the y-axis.
The straight line l_2 passes through the origin and has gradient 3. The lines l_1 and l_2 intersect at the point B.
c Calculate the coordinates of B.
d Calculate the exact area of $\triangle OAB$.

1.03 Coordinate Geometry: Linear Graphs

Chapter 5 · Original slide 42

Exercise 5G – Pure Y1 Page 102-103

Extension Problems

- [MAT 2001 1C]
  The shortest distance from the origin to the line $3x + 4y = 25$ is what?

